

## Trained



train



Large labelled  
training set

Corpus-specialized  
features

Standard



Cross-env.

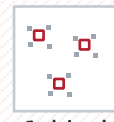


## Training-free, prior work



Frozen patch  
features

Standard



Codebook

Prior-selected / aggregated features

Cross-env.



Segmentation CLS attention



Segmentation CLS attention

## Ours



Frozen patch  
features

Standard



Within-image

Balanced, matchable patch evidence



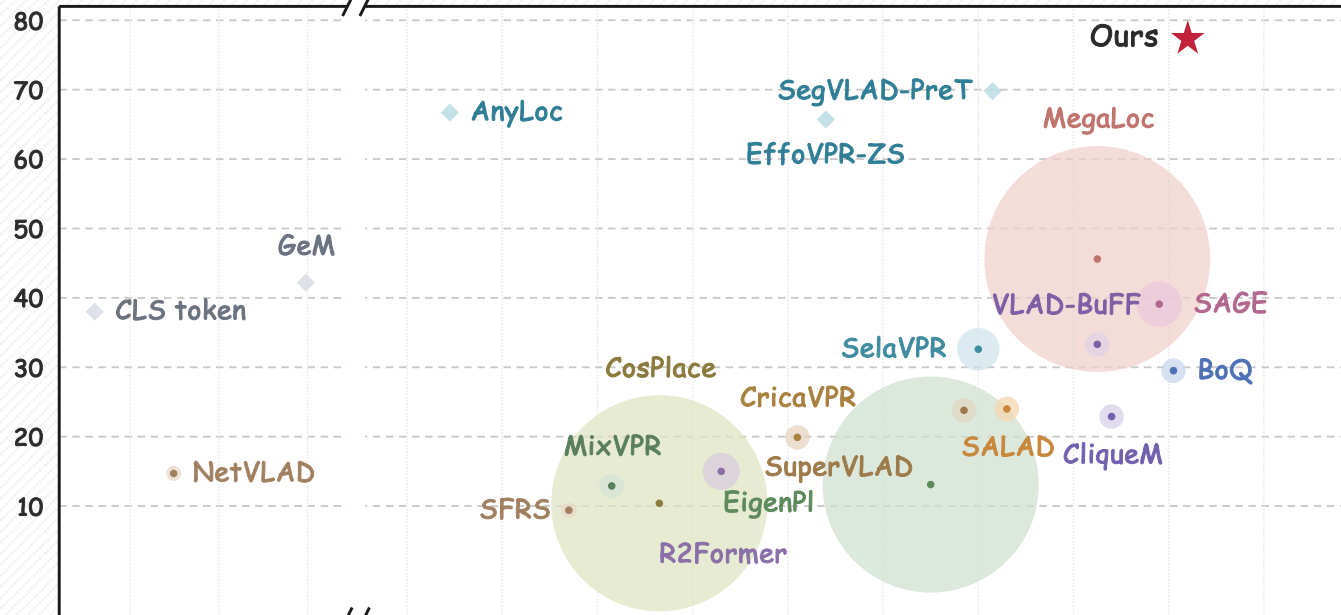
Across-image

Cross-env.



## Training cost vs. performance

VP-Air R@1 (cross-environment benchmark, %)



★ ours (training-free)

♦ training-free

● trained

● 1 million images

● 5 million images

## Performance across 17 benchmarks

